Cyber Security

Cybersecurity is not just for IT people, it's for **everyone**. If you use the internet, you need cybersecurity.

contents

- Week 1 – Introduction to Cybersecurity
- Fundamentals of cyber security
- History of cyber security
- What is Cybersecurity? Importance & Applications
- Types of Cyber Threats (malware, phishing, ransomware, DoS, etc.)
- Cybersecurity Terminology (threats, vulnerabilities, risks, exploits)
- CIA Triad (Confidentiality, Integrity, Availability)
- Careers in Cybersecurity

Pre-Activity Questions

- How many of you lock your phone with a PIN or fingerprint?
- Have you ever got a spam message or phishing email?
- What would happen if someone hacked your Instagram or bank account?

What is Cyber Security?

Definition:
Cybersecurity is the practice of protecting systems, networks, software, and data from digital attacks, theft, and damage.

It includes:

- **Computers** – Protection from malware/viruses.
- **Networks** – Preventing unauthorized access (like Wi-Fi hacking).
- **Applications** – Securing apps from bugs and attacks.
- **Data** – Keeping sensitive data safe from leaks.

Just like we lock our doors at home to protect from thieves, we use cybersecurity to protect our digital life from hackers.

Cybersecurity

Importance of Cybersecurity

"Why do we need cybersecurity? Because hackers are everywhere. If we don't protect ourselves..."

- **Personal protection** – Keeps our photos, emails, and bank accounts safe.
- **Business protection** – Prevents companies from losing money and data.
- **National protection** – Safeguards governments from cyber wars and digital spying.
- **Trust** – Without security, people won't trust online banking, shopping, or social media.

Example: *If Amazon or SBI got hacked tomorrow, would you feel safe using them again?*

Applications of Cybersecurity

"Cybersecurity is not only about hackers. It has **real applications** everywhere."

- **Banking & Finance** – Protecting online transactions (UPI, credit cards).
- **Social Media** – Protecting your Instagram, WhatsApp, Facebook accounts.
- **Healthcare** – Protecting patient data in hospitals.
- **E-commerce** – Safe shopping on Flipkart, Amazon.
- **Defence & Government** – Protecting sensitive data from cyber wars.
- **Engineering & IoT** – Securing smart devices (CCTV, smart homes, cars).

Activity: *"Think of one app you use daily (WhatsApp, Paytm, Swiggy) → How do you think cybersecurity protects it?"*

History of cyber security

- The first recorded cyber crime was recorded in year 1820.
- The first spam email took place in 1978 when i was sent over the Arpanet.
- The first Virus was installed on an Apple Computer in 1982.

Types of **cyber threats**

- **Phishing**
- **Malware Attacks**
- **Ransomware**
- **DoS and DDoS Attacks**
- **Insider Threats**
- **Spoofing**
- **Brute Force Attack**
- **Zero-day Attack**

Phishing

Meaning:
Phishing is a cyber attack where attackers send fake messages or emails pretending to be from trusted sources like a bank, government office to steal personal details like passwords or bank information.

How it Happens:
A fake email or SMS is sent with a link to a duplicate website. When the victim enters details, the attacker steals the information.

Example:
A person receives a message: "Congratulations! You won 10 lakh. Click here to claim."
Excited, they click the link and enter their bank details. Soon, all their money disappears.

Malware

Meaning:

Malware(malicious software) is a type of software that is created to harm a computer, steal data, or take control of a system.

How it Happens:

Attackers send malware through fake websites, free downloads, email attachments, or pop-ups. Once it is opened malware is installed, and starts damaging the system or collecting personal information.

Example:

A student downloads a free video editing tool from an unknown site. It installs malware that silently collects saved passwords and files.

Ransomware

Meaning:

Ransomware is a virus that locks the user's files or system and demands money (ransom) to unlock them.

How it Happens:

Attackers send an infected file through email. Once it is opened, the virus encrypts all files and shows a message demanding payment to recover data.

Example:

An office manager opens a PDF from email. All documents get locked, and a screen message says:

"Send ₹2.00,000 in Bitcoin to unlock your data."

DoS & DDoS Attacks

Meaning:

A DoS attack is when hackers overload a website or server with too many requests, making it slow or completely unavailable.

- A DDoS attack is similar to a DoS attack, but here, hackers use multiple devices to attack a website or server at the same time to cause a bigger disruption.
- How it Happens:

Dos uses one system, while DDoS uses many infected systems ms (botnets) to send continuous fake requests to a website, making it unavailable to real users.

- Example:

An online shopping website is attacked during a big sale event, causing the site to crash and preventing real customers from making purchases.

Zero-Day Attack

Meaning:

A zero-day attack happens when hackers find a hidden weakness in software before the company knows about it and use it to attack.

- How it Happens:

A newly launched software has a security

- bug. Hackers discover the flaw and create an attack before developers release a fix. Victims using the software become vulnerable.
- Example:
- A browser has a coding error. Hackers send a fake link that installs spyware. Company hasn't released any update yet.
- These attacks are very dangerous because there is no immediate solution available.

Insider Threat

Meaning:

An insider threat is caused by someone from within an organisation like an employee or ex-worker who misuses their access to cause harm.

- How it Happens:

The insider leaks sensitive files,introduces malware, or shares login credentials intentionally or by mistake.

- Example:

An employee downloads client data before resignation and gives it to a competitor company.

Spoofing

Meaning:

Spoofing is when an attacker pretends to be someone else to trick users or systems into giving access or information.

- How it Happens:

A fake email ID, phone number, or website is created that looks similar to the real one, and people are fooled into trusting it.

- Example:
- An email comes from "support@paypal.com" (with the number 1 instead of "l"). It asks for account verification and steals login details.

Brute Force Attack

Meaning:

Brute force attack is when an attacker tries many combinations of different username and password combinations repeatedly until they find the correct one for gaining access to an account or system.

- How it Happens:

The attacker runs a script that keeps trying different passwords.

- If the password is weak, the attacker gains access.
- The hacker can now steal data or misuse the account.
- Example: A social media account with the password "password193" is hacked within seconds using a brute force attack..

Home work

- Types of hackers
- And download fire fox
- https://www.firefox.com/en-US/?utm_campaign=SET_DEFAULT_BROWSER

Certifications of Cyber security

Types of Cyber Security certifications

- CompTIA IT Fundamentals (ITF+)
- CompTIA Network+
- CompTIA Security+
- CompTIA Cybersecurity Analyst (CySA+)
- CompTIA Cloud+
- EC-Council Certified Network Defender (CND)
- Cisco Certified CyberOps Associate
- GIAC Information Security Fundamentals (GISF)
- Microsoft Certified: Security, Compliance, and Identity Fundamentals (SC-900)

Types of Cyber Security certifications

- Certified Ethical Hacker (CEH)
- CompTIA Pentest+
- Certified in Risk and Information Systems Control (CRISC)
- Certified Information Systems Auditor (CISA)
- Certified Cloud Security Professional (CCSP)
- Certified Information Security Manager (CISM)

CompTIA IT Fundamentals (ITF+)

- ITF+ covers basic IT concepts such as hardware, software, networking, and security. It's used by beginners to build a foundational understanding of technology and serves as a stepping stone to more advanced IT roles.

Why it is required:

- ITF+ is essential for those new to the tech field, helping them get comfortable with technology before advancing to other certifications or IT roles.

CompTIA Network+

covers networking fundamentals, including network setup,management, and troubleshooting. It's used by network professionals to ensure efficient and secure network operations in IT environments.

Why it is required:

Network+ is needed for network administration and support roles, proving proficiency in managing and securing network systems.

CompTIA Security+

- Security+ covers cybersecurity basics, such as threat management, risk identification, and securing networks. It's used by professionals to protect systems from cyber threats and secure networks.

Why it is required:

- Security+ is required for cybersecurity roles, as it demonstrates the ability to safeguard data and systems from various cyber threats.

CompTIA Cybersecurity Analyst (CySA+)

- CySA+ focuses on proactive cybersecurity, such as threat detection and incident response. It's used by security analysts to monitor and respond to security events within networks.

Why it is required:

- CySA+ is required for specialized roles in cybersecurity, where professionals must actively monitor, detect, and respond to security threats.

CompTIA Cloud+

- Cloud+ focuses on cloud technologies, deployment, and security. It's used by cloud professionals to manage and secure cloud systems across various platforms like AWS and Azure.

Why it is required:

- Cloud+ is required for cloud administrators and engineers, ensuring proficiency in managing and securing cloud environments.

Microsoft Certified:

- Security, Compliance, and Identity Fundamentals (SC-900)
- SC-900 is a beginner-level certification that covers fundamental security, compliance, and identity concepts, focusing on Microsoft security solutions such as Azure Security, Microsoft 365, and identity protection.
- Why is it required?
- This certification helps beginners understand security, compliance, and identity in Microsoft environments, preparing them for advanced cloud security roles.

Certified Information Security Manager (CISM)

- CISM focuses on managing and governing information security programs. It's used by security managers to oversee risk management, incident response, and security governance.
- Why it is required:
- CISM is required for information security management roles, ensuring professionals can lead security programs and align them with business goals.

CISM

EC-Council Certified Network Defender (CND)

- CND teaches network defense techniques, including monitoring traffic and securing devices. It's used by network security professionals to safeguard an organization's network from cyber threats.
- Why it is required:
- CND is needed for network security professionals focused on defending against attacks and ensuring robust network security.

Cisco Certified CyberOps Associate

- Cisco CyberOps Associate covers security operations, monitoring, and incident response using Cisco technologies. It's used by security professionals to manage security operations in Cisco environments.
- Why it is required:
- Cisco CyberOps is required for security professionals in roles that involve monitoring and responding to security incidents in Cisco networks.

Certified Information Systems Auditor (CISA)

- CISA focuses on auditing IT systems to ensure security, compliance, and risk management. It's used by IT auditors to assess and improve the effectiveness of internal controls.
- Why it is required:
- CISA is required for auditing roles to ensure that systems are secure and compliant with industry standards and regulations.
- CIS

CompTIA Pentest+

- Pentest+ focuses on penetration testing, vulnerability assessments, and securing systems. It's used by professionals to perform ethical hacking and help organizations strengthen security.
- Why it is required:
- Pentest+ is necessary for penetration testing roles, demonstrating the skills to identify and fix vulnerabilities in systems and networks.

Certified Ethical Hacker (CEH)

- CEH focuses on ethical hacking techniques, including penetration testing and vulnerability assessments. It's used by professionals to identify and fix security weaknesses in systems.
- Why it is required:
- CEH is required for penetration testers and ethical hackers who are tasked with improving security by simulating cyberattacks.

GIAC Information Security Fundamentals (GISF)

- GISF introduces basic cybersecurity concepts like encryption and network security. It's used by beginners to get an overview of information security.

Why it is required:

- GISF is needed to establish a foundational understanding of cybersecurity for those new to the field, often as a prerequisite to advanced certifications.

Job Roles

Job Roles

- Ethical Hacker
- Cyber security analyst
- Security Engineer
- Security Analyst
- Soc analyst
- Penetration Tester
- Security Consultant
- Security auditor
- Network security Engineer
- Malware Analyst
- Vulnerability Assessor
- Network Administrator